Math 105L **Lab 2 Part 1**

Linear Models: Velocity & Acceleration in Free-Fall

1. Experimental Setup & Data Cleaning

An ultrasound motion detector records the height $h(t)$ (in meters) of a falling object every 0.03 seconds:

- **Data Cleaning:** For $t > 0.42$ s, sensor reflections cause erroneous rebounds where the object falsely appears to rise. These points are removed to isolate true unhindered free-fall on $[0, 0.42]$.
- **Average Velocity Computation:** For each interval $[t_k, t_{k+1}]$, the average rate of change of height is:

$$\mathbf{v_{avg}} = \frac{\mathbf{h(t_{k+1}) - h(t_k)}}{\mathbf{t_{k+1} - t_k}} \quad \text{(units: meters/second)}$$

- **Midpoint Assignment Rule:** To maximize numerical accuracy, the computed average velocity over $[t_k, t_{k+1}]$ is associated with the **center of the time interval**:

$$\mathbf{t_{mid}} = \frac{\mathbf{t_k + t_{k+1}}}{\mathbf{2}}$$

2. Classroom Demonstration on Blackboard: The First Time Step

Over the interval from $t = 0$ to $t = 0.06$ seconds ($h(0) = 1.542$ m, $h(0.06) = 1.527$ m):

- **Midpoint Time:** $t_{\text{mid}} = \frac{0+0.06}{2} = \mathbf{0.03}$ s.
- **Average Velocity:**

$$v_{\text{avg}} = \frac{1.527 - 1.542}{0.06 - 0} = \frac{-0.015}{0.06} = \mathbf{-0.25} \text{ m/s}$$

- Near $t = 0$, the height curve is almost horizontal, explaining why the initial velocity is small in magnitude.

The Velocity vs. Time Scatter Plot: A Linear Model When the calculated average velocities are plotted against the interval midpoints, the points align along a clear **straight line**:

$$\mathbf{v(t) = mt + b}$$

- **Negative Velocities:** Indicate downward directional motion ($h(t)$ is decreasing).
- **Velocities Becoming More Negative:** The object is speeding up downward.
- **Slope of the Velocity Line (m):** Represents the rate of change of velocity:

$$\text{Slope } \mathbf{m} = \frac{\mathbf{\Delta v}}{\mathbf{\Delta t}} = \text{Acceleration } \mathbf{a} \approx \mathbf{-9.8} \text{ m/s}^{\mathbf{2}}$$

3. Core Synthesis

- While the position graph $h(t)$ is **quadratic (curved downward)**, its derivative $v(t)$ is **linear**.
- The constant slope of this linear velocity model gives the physical constant of downward gravitational acceleration $g \approx 9.8$ m/s^2.